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Roll No

EE-5001-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Utilisation of Electrical Energy

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions,
ii) All questions carry equal marks.

1. a) What are polar curves? Explain Rousseau diagram and its importance in illuminating engineering.
b) A 100cp lamp is hung 2m above the centre of a circular area of 3 diameters. Determine the illumination at
 - i) The centre of area
 - ii) A point of the circumference of the area
 - iii) Average illumination.Find also the average illumination if a reflector of 50% efficiency is used.
2. a) List the methods of electric heating. Explain direct and indirect resistance heating in detail.
b) What is Induction heating? Explain high frequency induction furnace with its advantages.
3. a) Discuss the principle of arc welding and differentiate carbon and metallic arc welding.
b) What is electroplating and what for is it done? Describe the various operations involved in electroplating.

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4.
 - a) Explain speed - time curves for electric traction.
 - b) The distance between two stations is 1 km and the schedule speed is 30 km/h. Station stopping time 20 sec. Assume braking retardation 3 km/h/s and maximum speed 1.25 times the average speed. Determine the acceleration required to run the service if speed-time curve is trapezoidal curve.
5.
 - a) Explain the features of a modern single phase 25 kV a.c traction system.
 - b) Explain various methods of speed control of traction motors.
6.
 - a) Derive an expression for the temperature rise of an electric motor. State the assumptions made.
 - b) Explain the mechanism of train movement.
7.
 - a) What is an electric drive? List the major parts of an electrical drive. State the advantages and disadvantages of electrical drive.
 - b) Define air conditioning. On what factors does the air-conditioning depend? Explain in detail.
8. Write short notes on (any two)
 - a) Function of fly wheel in rolling mill drives
 - b) Inverse square and cosine law
 - c) Arc furnaces transformer
 - d) Adhesive weight and coefficient of adhesion
 - e) Modern 25kV Ac single phase traction systems.
